Part 3 - Gradient Descent by hand
(manually done)

Given:

$m = [m_1, m_2] = [-1, 2]$ $b = 1$

$x = [(1,3), (4,10)] \rightarrow x_1 = (1,4), x_2 = (3,10)$

$y = (5,6)$

Learning rate $\eta = 0.01$

Group of 1 → 1 full iteration shown

Step 1: Prediction ($\hat{y} = x \cdot m + b$)

$\hat{y}_1 = (1)(-1) + (3)(2) + 1 = -1 + 6 + 1 = 6$

$\hat{y}_2 = (4)(-1) + (10)(2) + 1 = -4 + 20 + 1 = 17$

$y = (6, 17)$

Step 2: Cost (MSE), $n = 2$

$e_1 = \hat{y}_1 - y_1 = 6 - 5 = 1$

$e_2 = \hat{y}_2 - y_2 = 17 - 6 = 11$

$J = (1/n) \cdot \sum e_i^2 = \left(\frac{1}{2}\right)(1^2 + 11^2) +$

$\left(\frac{1}{2}\right)(1 + 121) =$

$= \left(\frac{1}{2}\right)(122) =$

$= 61$

Step 3: Chain rule

Link 1: $\partial J / \partial \hat{y}_i = \left(\frac{2}{n}\right) \cdot e_i$

$= \left(\frac{2}{2}\right) \cdot e_i$

$= e_i$

$\Rightarrow \frac{\partial J}{\partial \hat{y}_1} = 1 \quad \frac{\partial J}{\partial \hat{y}_2} = 11$

Link 2a: $\dfrac{\partial \hat{y}_i}{\partial m} = x_i$

(since $\hat{y}_i = x_i \cdot m + b$)

Chain rule combined for m:

$$\frac{\partial J}{\partial m_1} = \frac{\partial J}{\partial \hat{y}_1} \cdot x1_1 + \frac{\partial J}{\partial \hat{y}_2} \cdot x1_2$$

$= (1)(1) + (11)(4)$

$= 1 + 44$

$= 45$

$$\frac{\partial J}{\partial m_2} = \frac{\partial J}{\partial \hat{y}_1} \cdot x2_1 + \frac{\partial J}{\partial \hat{y}_2} \cdot x2_2$$

$= (1)(3) + (11)(10)$

$= 3 + 110$

$= 113$

Link 2b: $\dfrac{\partial \hat{y}_i}{\partial b} = 1$

Chain rule combined for b:

$$\frac{\partial J}{\partial b} = \frac{\partial J}{\partial \hat{y}_1} \cdot 1 + \frac{\partial J}{\partial \hat{y}_2} \cdot 1$$

$= 1 + 11 = 12$

$\nabla J = (45, 113, 12)$

Step 4: Update ($\eta = 0.01$)

m1_new $= -1 - (0.01)(4.5)$

$= -1 - 0.045$

$= -1.45$

m2_new $= 2 - (0.01)(113)$

$= 2 - 1.13$

$= 0.87$

$b\_new = 1 - (0.01)(12)$

$= 1 - 0.12$

$= 0.88$

Step 5: Recheck cost

$\hat{y}_1\_new = (1)(-1.45) + (3)(0.87) + 0.88$

$= -1.45 + 2.61 + 0.88$

$= 2.04$

$\hat{y}_2\_new = (4)(-1.45) + (10)(0.87) + 0.88$

$= -5.80 + 8.70 + 0.88$

$= 3.78$

$e_1 = 2.04 - 5 = -2.96$

$e_2 = 3.78 - 6 = -2.22$

$J\_new = \left(\frac{1}{2}\right)\left((-2.96)^2 + (-2.22)^2\right)$

$= \left(\frac{1}{2}\right)(8.7616 + 4.9284)$

$= \left(\frac{1}{2}\right)(13.69)$

$= 6.845$

Table	m_1	m_2	b	J
iter 0	-1	2	1	61
iter 1	-1.45	0.87	0.88	6.848

Trend: cost dropped $61 \to 6.848$ in one step ($\simeq 89\%$)

m_2 moved the most since x_2's inputs (3, 10) are largest, so its gradient (113) is largest.

The chain rule amplifies whichever input has more magnitude.